# PV102 Day 3: Load Profile and Energy Budget

Students turn a client wish list into a defensible load table, daily watt-hour estimate, priority list, and unknowns register.

PV102 | CTS Day 19 | Instructor Run Sheet
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

## Core rule

Watts are moments. Watt-hours are work over time.

## Learning targets

- Distinguish watts from watt-hours.
- Calculate daily energy use from power and time.
- Rank loads by necessity, runtime, surge risk, and design impact.
- Explain how load decisions affect PV, battery, inverter, and budget.

## Readiness and materials

| Before students arrive | Materials |
|---|---|
| Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled. | Calculators<br>Load cards<br>Appliance nameplate examples<br>Student load table<br>Chart paper<br>Projector and slide deck |

## Five-hour run of show

| Time | Block | Instructor move |
|---|---|---|
| 10:00-10:15 | Opening and continuity | Connect Day 2 to today's rule. Name what students already know and what new judgment is being added. |
| 10:15-10:40 | Morning quiz and review | Students work independently for 12-15 minutes. Review by asking for evidence, not just letters. |
| 10:40-11:25 | Concept lesson | Use slides 1-14. Pause after each technical rule and ask what decision it changes. |
| 11:25-12:15 | Guided practice | Model one example on the board. Students complete a quick pair-check before lunch. |
| 12:15-12:45 | Lunch/reset | Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure. |
| 12:45-1:00 | Lab brief | Assign teams and roles. Review acceptance criteria and stop-work triggers. |

| | | |
|---|---|---|
| 1:00-2:15 | Client Load Interview and Energy Budget | Teams interview a mock client, build a daily load table, and recommend one load reduction. |
| 2:15-2:40 | Handoffs and peer verification | Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns. |
| 2:40-3:00 | Exit ticket and reset | Collect packets. Preview next step: PV102 Day 4 uses load and site information to explore module behavior, strings, and shade impact. |

## Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use load profile and energy budget to make the next design decision safer and more defensible. The rule is simple: Watts are moments. Watt-hours are work over time."

## Teaching notes

| Move | Look-for |
|---|---|
| Ask | What decision does this evidence change? |
| Listen for | Students separating facts, assumptions, and unknowns. |
| Correct gently | Component-first answers. Redirect to job, evidence, boundary, then component. |
| Require | A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system. |

## Lab instructor procedure

- Choose one scenario and list all loads.
- For each load, record watts, hours per day, daily Wh, priority, and unknowns.
- Mark surge or motor loads.
- Calculate total daily Wh and essential daily Wh.
- Recommend one load change that improves system practicality without weakening the client need.
- Prepare a handoff showing the load table and top three unknowns.

## Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

## Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

## Day handoff

PV102 Day 4 uses load and site information to explore module behavior, strings, and shade impact.

## Instructional sources


- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.